GRADE 7 ACCELERATED · UNIT 4 · LESSONS 7–8

Real-World Order of Operations

Question Bank 4.07 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, substitute and evaluate.

1. $4x^2$ when $x=-3$

36

$4(-3)^2=4(9)$.

2. $\frac{1}{2}mv^2$ when $m=50$ and $v=6$

900

$25(36)$.

3. $\frac{a+b^2}{c}$ when $a=6$, $b=5$, $c=2$

$\frac{31}{2}$

The bar groups the whole numerator: $6+25=31$.

4. $2(x-y)^2$ when $x=3$ and $y=7$

32

$2(-4)^2=2(16)$.

For questions 5 and 6, write and evaluate an expression. Give the unit.

5. A square room of side 11 feet is floored at 3.20 dollars a square foot, with a flat 75 dollars for delivery. Find the total cost.

$11^2(3.20)+75=462.20$ ***dollars***

The rate is per square foot, so it multiplies the area; the delivery is flat, so it does not.

6. A rectangular deck 14 feet by 9 feet is stained at 1.15 dollars a square foot, with a flat 48 dollars for equipment hire. Find the total cost.

$14(9)(1.15)+48=192.90$ ***dollars***

The area is 126 square feet.

Two formulas.

7. Body mass index is $B = \frac{703w}{h^2}$, where w is a weight in pounds and h is a height in inches. Find B to the nearest tenth.

a. $w = 150$, $h = 65$

25.0

$\frac{105{,}450}{4{,}225} = 24.958\ldots$

b. $w = 180$, $h = 70$

25.8

$\frac{126{,}540}{4{,}900} = 25.824\ldots$

c. $w = 200$, $h = 72$

27.1

$\frac{140{,}600}{5{,}184} = 27.122\ldots$

8. Kinetic energy is $K = \frac{1}{2}mv^2$, for mass m in kilograms and speed v in meters per second. Find K in joules.

a. $m = 1{,}200$, $v = 15$

135,000 joules

b. $m = 850$, $v = 12$

61,200 joules

c. $m = 70$, $v = 8$

2,240 joules

Working backwards.

9. Find the missing value in each situation.

a. $\frac{1}{2}mv^2 = 900$ and $v = 6$. Find m.

m = 50

$\frac{1}{2}m(36) = 900$, *so* $18m = 900$.

b. $x^2 + 5 = 41$, where x is positive. Find x.

x = 6

c. A square lawn of side n feet is turfed at 2.50 dollars a square foot with a flat 40 dollars delivery, and the total is 2,290 dollars. Find n.

n = 30

$n^2(2.50) = 2{,}250$, *so* $n^2 = 900$.

Reading the situation.

10. Two problems use the numbers 20, 3 and 5. **A:** a gym charges 20 dollars to join plus 3 dollars for each of 5 classes. **B:** a gym charges 20 dollars for each of 3 months, plus 5 dollars to join. Find both totals, and explain why the words 'plus' and 'each' do not decide the operation.

A is $20+3(5)=35$ ***dollars. B is*** $20(3)+5=65$ ***dollars. Both problems contain the same numbers and the same two words, so the words cannot be what decides; what decides is which amount is repeated.***

This is the state guide's named misconception, tested directly: circling the numbers and boxing the key words produces identical markings on two problems with different answers.

11. A square yard of side 15 feet is turfed at 4.80 dollars a square foot, with a flat 120 dollars for delivery. Which expression gives the total cost?

- ○ A. $15(4.80)+120$
- ● **B.** $15^2(4.80)+120$
- ○ C. $15^2(4.80+120)$
- ○ D. $15^2+4.80+120$

B

12. Select ALL of the expressions whose value is 45 when $x=3$.

- ☐ A. $(5x)^2$
- ☒ **B.** $\frac{135}{x}$
- ☒ **C.** $5x^2$
- ☒ **D.** $3x^2+18$
- ☐ E. $5x\cdot 2$
- ☒ **F.** x^2+36

B, C, D and F